

Amazon CloudWatch Overview (Solutions Architect Focus)

What is Amazon CloudWatch?

Amazon CloudWatch is a **monitoring and observability service** built for DevOps engineers, developers, site reliability engineers (SREs), and architects. It collects monitoring and operational data in the form of **logs, metrics, and events** from AWS resources, applications, and on-premises servers.

Key Use Cases

- Monitor **EC2 CPU**, memory, disk, network utilization
 - Track **Lambda function invocations, durations, and errors**
 - Set **alarms for Auto Scaling**, billing thresholds, etc.
 - Visualize metrics in **CloudWatch Dashboards**
 - **Trigger actions** (e.g., reboot EC2) using alarms
 - Analyze **application logs** for troubleshooting
-

Core Components

1. Metrics

- Numerical data points over time (e.g., CPUUtilization)
- Namespace-based (e.g., [AWS/EC2](#), [AWS/Lambda](#))

- Custom Metrics can be published from applications or on-prem systems

2. Alarms

- Used to **automate actions** based on thresholds (e.g., if CPU > 80% for 5 min)
- Can trigger:
 - **SNS notifications**
 - **Auto Scaling**
 - **EC2 actions** (stop, terminate, reboot)

3. Logs

- CloudWatch Logs lets you **store, monitor, and analyze logs**
- Common sources:
 - **EC2** via CloudWatch Agent
 - **Lambda function logs**
 - **API Gateway logs**
- Supports **log retention, search, metric filters**

4. Dashboards

- Customizable visual representation of metrics
- Display data from **multiple services/accounts/regions**
- Great for operational monitoring and executive reporting

5. Events / EventBridge

- CloudWatch Events is now part of **Amazon EventBridge**
 - Enables **event-driven architecture**
 - Automatically responds to changes in your environment (e.g., EC2 state change, CodeDeploy failures)
-

Architectural Best Practices

- Use **CloudWatch Agent** for detailed EC2 instance metrics (e.g., memory, disk I/O)
 - **Aggregate logs** from all regions to a centralized account for security auditing
 - Integrate **CloudWatch Logs with Lambda** for serverless monitoring pipelines
 - Create **alarms** for **billing metrics** to control unexpected costs
 - Use **CloudWatch Insights** to analyze logs at scale with SQL-like queries
-

Pricing (High Level)

- **Basic monitoring (EC2):** Free (5-min intervals)
 - **Detailed monitoring:** Charged (1-min intervals)
 - **Logs ingestion, storage, and retrieval:** Priced per GB
 - **Custom metrics, dashboards, alarms:** Priced per usage
-

Exam Tips (SAA-C03)

- CloudWatch **does not** automatically monitor memory/disk space on EC2 — you must install the **CloudWatch Agent**
- Use **CloudWatch Logs + Lambda** to create serverless remediation
- **CloudWatch Alarms** can trigger **Auto Scaling actions**
- To **monitor application logs**, use **CloudWatch Logs Insights**
- **EventBridge** supports **third-party SaaS integration** and more complex event routing